AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2025

Business Mathematics Paper II

Time: 1 hour 5 minutes Marks: 20

Annual Examinations 2025 AKU-EB for Teaching & Learning only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are SIX questions. Answer ALL the questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.

6. You may use a scientific calculator if you wish.

List of Formulae

Note:

- The symbols used in the formulae have their usual meanings.

Interest and Annuities

$$I = \frac{PTR}{100} \qquad A = P \times \left(1 + \frac{r}{n}\right)^{nt} \qquad R = \left(1 + \frac{i}{n}\right)^{n} - 1$$

$$j = \left(1 + \frac{r}{m}\right)^{m} - 1 \qquad A = \left(\frac{(1+i)^{n} - 1}{i}\right) \qquad \text{Annuity} = R \times \left(\frac{(1+i)^{n} - 1}{i}\right)$$

Linear Equations, Functions and their Graphs

$$ax + by + c = 0 \qquad \frac{x}{a} + \frac{y}{b} = 1 \quad a \neq 0 \text{ and } b \neq 0$$

$$m = \frac{y_n - y_m}{x_n - x_m} \qquad y - y_m = m(x - x_m)$$

Quadratic Equations, Functions and their Graphs

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad y = a(x-h)^2 + k$$

Derivative of Algebraic Functions

$$\frac{d}{dx}[c] = 0 \qquad \frac{d}{dx} x^n = n \times x^{n-1}$$

$$\frac{d}{dx}[f(x)]^n = n[f(x)]^{n-1} \times f'(x) \qquad \frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x) \qquad \frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

Sequence and Series

$$a_n = a_1 + (n-1)d \qquad S_n = \frac{a_1(1-r^n)}{1-r}, \text{ if } |r| < 1 \qquad S_n = \frac{a_1(r^n - 1)}{r-1}, \text{ if } |r| > 1$$

$$S_n = \frac{n}{2}(2a_1 + (n-1)d) \qquad a_n = a_1 r^{n-1}$$

Matrices and Determinants

$$A^{-1} = \frac{1}{|A|} AdjA \qquad X = A^{-1}B \qquad X = \frac{|A_x|}{|A|}$$

AKU-EB Annual Examinations 2025 For Teaching & Learning only

Q.1. (Total 4 Marks)

The relationship between three quantities x, y and z is that z varies directly with x^2 and inversely with y.

Their relationship is shown in the given table.

x	0	1	2
y	1	1	3
z	0	3	p

Using the given information,

i. prove that $z = \frac{3x^2}{y}$. (3 Marks)

ii. find the value of p. (1 Mark)

AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 4 Marks)

a. Nisar invested Rs 500,000 in a savings account with a fixed annual interest rate of 7% compounded quarterly. Find the accumulated amount after 10 years.

b. Zeeshan deposited Rs 8,000 at the end of each period of six months for 9 years at 5% interest compounded semi-annually. Find his accumulated amount.

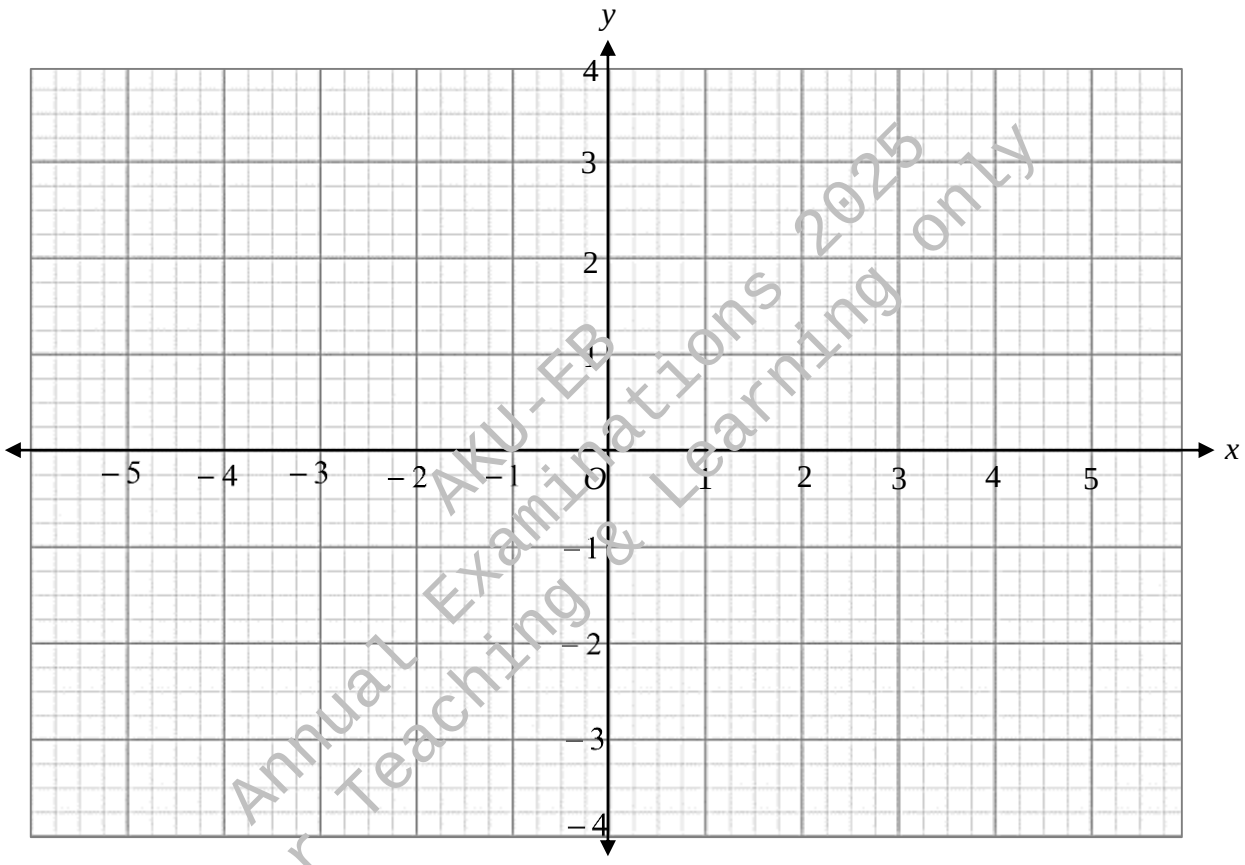
AKU-EB Annual Examinations 2025 for Teaching & Learning only

Q.3. (Total 3 Marks)

The given table contains the points satisfying the function.

x	−1	0	1
y	1	0	1

i. Using the given table, sketch the graph of the function. (2 Marks)



AKU-EB Annual Examinations 2025 for Teaching & Learning only

ii. On the graph, identify the vertex. (1 Mark)

(ATTEMPT EITHER PART a OR PART b OF Q.4.)

Q.4. (Total 3 Marks)

a. Find the solution of the following simultaneous linear equations.

$2x + 3y = 6$

$4x - y = 5$

AKU-EB Annual Examinations 2025 for Teaching & Learning only

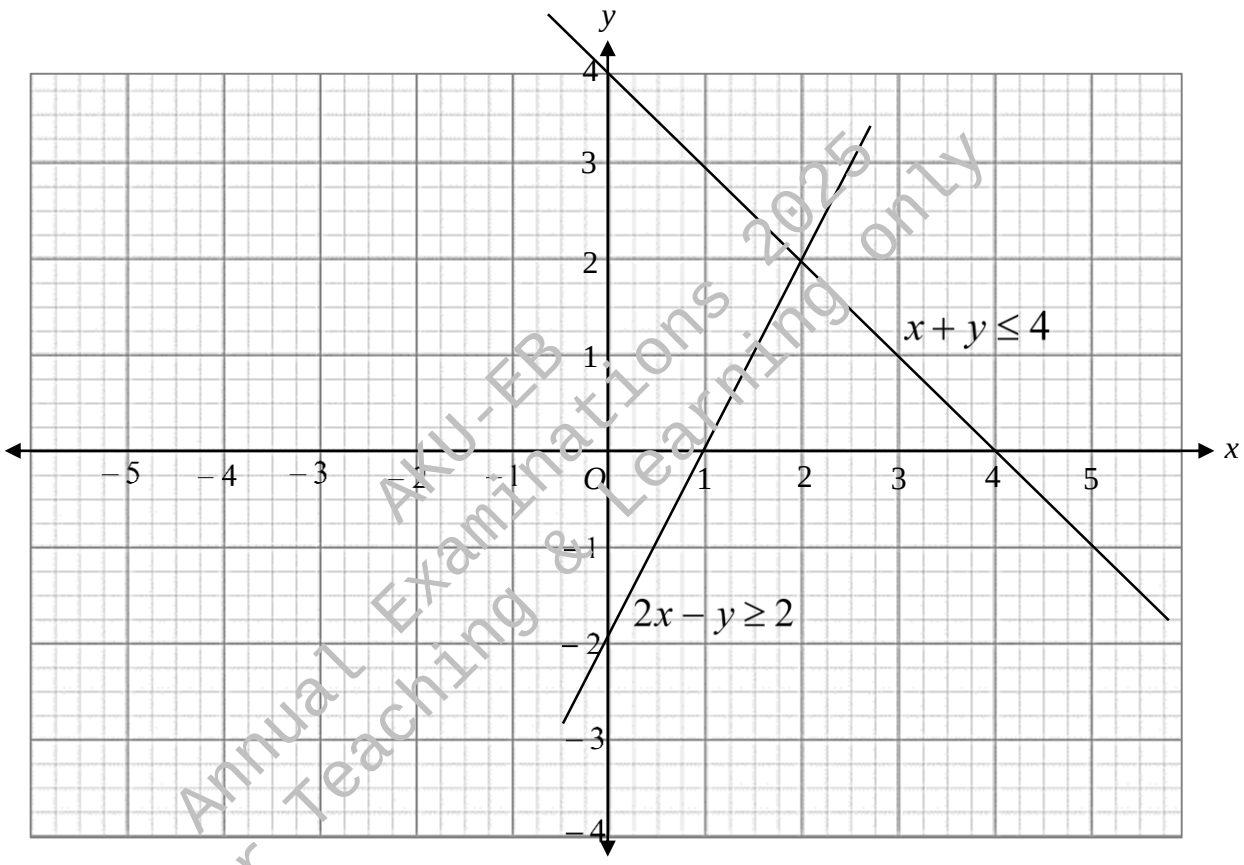
(ATTEMPT EITHER PART a OR PART b OF Q.4.)

b. Consider the given linear programming problem.

$x + y \leq 4 \qquad 2x - y \geq 2 \qquad x \geq 0,\ y \geq 0$

On the given graph,

i. shade the feasible region. (1 Mark)



AKU-EB Annual Examinations 2025 for Teaching & Learning only

ii. identify all the corner points of the feasible region. (2 Marks)

_____

_____

Q.5. (Total 3 Marks)

Find the values of x and y by solving the given matrix equation using Cramer's rule.

$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 3 Marks)

a. The sum of 50 terms of an arithmetic sequence is 3,725. If the common difference of the series is 3, then find the

i. first term of the series. (2 Marks)

ii. fifteenth term of the series. (1 Mark)

AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

b. Aslam invested Rs 300,000 in a business. The capital grew at a growth rate of 9.04% in 10 years. Calculate the total worth of the capital after 10 years. (3 Marks)

AKU-EB Annual Examinations 2025 for Teaching & Learning only

END OF PAPER

Please use this page for rough work

AKU-EB
Annual Examinations 2025
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2025
for Teaching & Learning only